TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Davide Ziviani	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	585 Purdue Mall, West Lafayette, IN 47907-2088			

2. Technology Overview

Title of Proposed Technology	**Development and Commercialization of a ±25 K/min Class Rapid Thermal Cycling Chamber**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		*Electrical · Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	*Environmental Measuring Instruments*	*Batteries*	*Manufacturing and Measurement/Evaluation Equipment*
Concept of Proposed Technology	**Core Concept** The Rapid Thermal Cycle Chamber proposed in this project is a next-generation environmental reliability testing system capable of achieving a high-speed temperature ramp rate of ±25 K/min across a wide range from -70 °C to +180 °C. This technology aims to secure a domestically developed (South Korean) platform by integrating heating, refrigeration, airflow, and control systems, moving beyond simple assembly to achieve full system localization. ● High-Performance Thermal Design: Optimization of high-output heating systems and two-stage (cascade) refrigeration structures to realize ±25 K/min ramp rates. ● Precision Airflow Engineering: Optimization of air circulation (ducts, fans, heat exchangers) based on CFD (Computational Fluid Dynamics) to maintain temperature uniformity within ±2 °C even during high-speed transitions.			

(max 10 pages)

- Standard-Compliant Control Logic: Internalization of control algorithms and test scenarios that meet international standards, such as IEC 60068-2-14.
- Eco-friendly & High Efficiency: Implementation of low-GWP (Global Warming Potential) refrigerants and energy-optimized operation strategies.
- Specimen-Centric Control: Integration of Tc (Test-cell) control options based on specimen temperature rather than just ambient air temperature to meet high-end ESS and R&D demands.

Key Functions and System Configuration

1) Performance
   - Temp. Range: -70 °C to +180 °C
   - Ramp Rate: ±25 K/min
   - Uniformity: ≤ ±2 °C / Accuracy: ±0.5 °C
2) Thermal System
   - High-output electric heaters
   - Cascade refrigeration system for rapid cooling
   - Anti-condensation and frost control for stable operation
3) Control System
   - Localization of HMI/PLC-based control systems
   - Recipe management, real-time monitoring, and data logging
   - Remote control and fault alarm systems
4) Smart Features
   - Predictive Maintenance based on equipment status data
   - Modular structure for expansion (e.g., humidity control, combined environments)
   -

Application Fields and Commercialization Potential

This technology serves as a critical infrastructure for high-value industries requiring extreme reliability:

- **Semiconductors & Packaging**: Reliability assessment of solder joints, package interfaces, and substrates.
- **Display & Optical Modules**: Evaluation of thermal fatigue, delamination, and thermal shock durability.
- **Automotive Electronics & Batteries**: Durability verification for **ECUs, ADAS sensors**, and power semiconductors.
- **Aerospace & Defense**: High-reliability component validation through repetitive extreme temperature testing.

Rationale for Joint R&D Project

Critical Technical Barriers

The development of a ±25 K/min high-speed thermal cycle chamber is a complex systems integration project that goes beyond simple equipment manufacturing. It requires the seamless

convergence of multi-layered technical elements:

- **High-Performance Thermal Engineering**: Design of high-output heating and multi-stage refrigeration systems for rapid ramp rates.
- **Precision Stability**: Ensuring stable temperature uniformity and control accuracy under high-speed thermal transitions.
- **Global Compliance**: Establishing test scenarios and verification systems that meet international environmental standards (e.g., IEC).
- **Industrial Application**: Integrating **Environmental Stress Screening (ESS)** operational expertise required by global semiconductor and automotive sectors.

Given these challenges, it is practically impossible for a single entity to master the entire spectrum of theory, design, manufacturing, and international standardization within a short timeframe.

Why Global Collaboration?

Participation of international research institutions is a critical requirement for securing global-tier competitiveness:

1) **Sharing Global Standards and Operational Expertise**: International partners provide practical know-how that is difficult to secure domestically, including:
    - Advanced test condition settings and realistic benchmarks for ramp performance.
    - Implementation of **Tc (Test-cell) control** requirements based on actual specimen temperatures.
2) **Securing Export Readiness:** The final goal is to penetrate the global market. By collaborating with international partners from the initial design phase, we ensure compliance with IEC standards and global customer requirements, securing export-ready technical specifications upon project completion.
3) **Benchmarking and Performance Validation:** Establishing an objective validation system through comparative evaluations with leading global equipment and international test protocols.

This collaborative R&D framework combines the Global Benchmarks & Standards of international partners with the Technical Implementation & Localization lead by domestic institutions.

1) **Strategic Goal:** Rapidly securing a "K-Rapid Chamber Platform" that satisfies global-tier specifications.
2) **Expected Impact:** This partnership transcends simple research; it creates a sustainable industrial ripple effect, including:
    - Successful Import Substitution of high-end equipment.
    - High-barrier market entry into the premium testing equipment sector.

Establishment of long-term export-strategic equipment portfolios.

Economic Impact

By securing core technologies for the ±25 K/min class market—currently dominated by global

(max 10 pages)

	leaders like ESPEC—this project will facilitate import substitution and establish a competitive edge for the global export market. Furthermore, it transitions from simple equipment supply to data-driven value creation through smart operation technologies.
Key Words	1. Rapid Thermal Cycling Technology 2. High-Ramp-Rate Environmental Chamber Localization 3. IEC-Compliant Reliability Testing Platform 4. Integrated High-Power Heating & Cascade Refrigeration

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	**R&D Objectives** The primary objective of this project is to develop an indigenous Rapid Thermal Cycle Chamber that simultaneously achieves a high-speed temperature ramp rate of ±25 K/min and a wide-range control span of -70 °C to +180 °C. By realizing premium environmental reliability testing equipment—currently dominated by global leaders—through domestic technology, we aim to establish a foundation for import substitution and secure technical competitiveness as an export-oriented testing platform compliant with international standards. **Detailed Objectives** 1) Integrated System Internalization ● Beyond simple manufacturing, the project focuses on the integrated design and internalization of core technologies that dictate rapid thermal cycle performance, including high-output heating, multi-stage refrigeration, precision air circulation, and advanced control systems, to ensure equipment stability and reliability. 2) Global Compliance & Market Readiness ● Through collaboration with international research partners, we will establish a design and verification framework that reflects global testing specifications and operational expertise. This ensures that international standard responsiveness and overseas market applicability are secured from the early development stage. 3) Modular Platform & Long-term Sovereignty ● As a mid-to-long-term goal, we aim to establish a modular and platform-based chamber architecture that can be expanded into humidity control and combined environmental testing. This will lead to complete technical self-reliance in premium environmental testing equipment, supporting high-reliability industries such as semiconductors, displays, and automotive electronics.			
R&D Target Specifications	1) Thermal Performance ● Rapid Ramp Rate: Achieving a high-speed transition of ±25 K/min or higher to drastically reduce testing time. ● Wide Operating Range: Precise control across the full spectrum from extreme cold (-70 °C) to high heat (+180 °C). 2) Thermal Quality and Control Precision ● Uniformity: Ensuring temperature deviation within the chamber remains below ±2.0 °C, even during rapid ramp-up/down phases.			

	● High Precision: Maintaining a steady-state control accuracy of ±0.5 °C to ensure the reproducibility of test results. 3) System Stability and Reliability ● Cyclic Durability: Ensuring mechanical and thermal stability during repetitive, high-stress thermal cycles. ● Environmental Management: Implementing advanced logic to suppress condensation and frost formation during the rapid cooling phase, preventing specimen damage. ● Long-term Operation: Securing system reliability for continuous, unattended long-term testing. 4) Control and Operation Features ● Localized Control: Integration of an indigenous HMI/PLC-based control system for technical independence. ● Smart Operation: Features include comprehensive recipe management, real-time status monitoring, and secure data logging. ● Remote Diagnostics: Support for remote control and automated fault diagnosis (Health Monitoring). 5) Scalability and Global Compliance ● International Standards: Pre-configured test scenarios complying with IEC 60068-2-14 and other global environmental test standards. ● Future-proof Architecture: A modular structural design that allows for future integration of humidity control and combined environmental stress (vibration/thermal) testing.
R&D Plan	**Research Approach** This project follows a three-phase strategic approach to ensure the transition from core technology acquisition to global-tier system validation. 1) Phase 1: Core Technology Design & Simulation-based Verification ● Individual design and analysis of high-output heating, cascade refrigeration, and airflow structures. 2) Phase 2: System Integration & Prototyping ● Assembly of the rapid thermal cycle chamber prototype and integration of localized control systems. 3) Phase 3: Performance Validation & Global Compliance Collaborative verification with international partners to ensure compliance with global standards and operational reliability. **Methodologies & Milestones** 1) Methodologies ● Systems Engineering (SE) based holistic design. ● Computational Fluid Dynamics (CFD) for structural flow optimization. ● Advanced Control Theory for high-speed thermal response. ● International Standards (IEC) compliance for test scenario validation.

2) Expected Milestones

- **M1**: Completion of core element technology design.
- M2: Full implementation of HMI/PLC and control systems.
- M3: Completion of the ±25 K/min rapid chamber prototype.
- M4: Verified achievement of ±25 K/min and -70 °C to +180 °C performance.
- M5: Final validation of international standard compliance and system optimization.

R&D Timeline (36 Months)

Phase	Duration	Primary Focus
Phase 1	Months 0–12	Core design, simulation, and basic control logic development.
Phase 2	Months 13–24	System integration, prototype manufacturing, and initial validation.
Phase 3	Months 25–36	Global joint verification, standard-compliant testing, and final optimization.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1) Requested Technical Contributions ● Global Operational Benchmarking: Provision of international operational standards and design requirements for ±25 K/min $ high-speed thermal cycle testing. ● Standard-based Methodology: Development of test scenarios and verification methodologies based on international environmental standards, such as IEC. ● Advanced ESS Concepts: Technical advisory on advanced Environmental Stress Screening (ESS) operations, specifically focusing on Tc (Test-cell) control based on specimen temperature. 2) Research Scope ● Review of Global Standards: Evaluation of international test conditions and performance metrics to ensure global competitiveness. ● Joint Scenario Development: Collaborative establishment of test scenarios for verifying international standard compliance. ● Analysis & Optimization: Interpretation of performance validation results and proposing strategic improvements for the final system. 3) Provided Resources ● Technical Documentation: Access to comprehensive technical data and literature regarding international standard-compliant testing. ● Expert Manpower: Involvement of high-level research personnel with extensive experience in high-speed thermal cycle operations. ● Technical Advisory Framework: A structured system for continuous consultation and joint performance verification. 4) Planned Key Tasks ● Requirement Definition: Defining technical requirements and providing design-phase consultancy to align with global market needs. ● Joint Performance Evaluation: Participation in the collaborative assessment during the prototype validation phase. ● Global Benchmarking: Reviewing performance levels against international standards and suggesting iterative enhancements.
Expected R&R of Korean SME Partner	1) Requested Technical Contributions ● Defining system architecture and overseeing the integrated design of the rapid thermal cycle chamber. ● Deriving performance specifications based on actual industrial requirements. ● Ensuring system stability within industrial environments through prototype fabrication. 2) Research Scope ● Structural design and comprehensive system integration. ● Prototype assembly and industrial environment testing. ● Review of cost-efficiency, mass-producibility, and maintainability. 3) Provided Resources

- Production facilities for equipment manufacturing and assembly.
- Dedicated technical workforce and a robust project management system.
- Extensive experience in testing equipment manufacturing and commercialization.

4) Planned Key Tasks
- Fabrication and integration of the high-speed rapid chamber prototype.
- Design refinement based on performance validation results.
- Establishment of commercialization strategies and planning for follow-up models.

Collaborative Research Institution / Academia

1) Requested Technical Contributions
- Research on thermal/refrigeration system design to achieve ±25 K/min ramp rates.
- CFD-based air circulation design to ensure temperature uniformity within the chamber.
- Development of advanced control algorithms for high-speed ramping and stabilization.

2) Research Scope
- Study on high-output heaters and two-stage (cascade) refrigeration configurations.
- Development of control logic (PID-based high-speed temperature control).
- Performance testing and empirical data analysis.

3) Provided Resources
- Thermal-fluid analysis software (CFD) and testing infrastructure.
- Environmental testing and measurement instrumentation.
- Dedicated research personnel (Professors, Research Fellows, and Graduate Students).

4) Planned Key Tasks
- Design and performance verification of core element technologies.
- Support for prototype performance evaluation and deriving technical improvement plans.
- Joint technical reviews and system optimization in collaboration with the lead institution.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** ● Acquisition of Core Rapid Chamber Technology: Securing indigenous core technologies—high-output heating, multi-stage refrigeration, airflow circulation, and precision control—that satisfy both the ±25 K/min ramp rate and the -70 °C to +180 °C temperature range. ● Implementation of a Global-Standard Testing Platform: Achieving equipment reliability ready for immediate field use by internalizing control logic and test scenarios compliant with international standards such as IEC 60068-2-14. ● Enhanced Reliability via International Validation: Ensuring objective performance credibility and technical maturity by applying global-tier operational expertise and verification methodologies through collaborative R&D with Purdue University. ● Foundation for Scalable & Platform-based Technology: Establishing a modular architecture that allows expansion into humidity control, combined stress testing, and Tc (Test-cell) control, facilitating evolution into next-generation ESS equipment. **Commercialization Potential** ● Commercialization of Import-Substitution Equipment: Providing a viable domestic alternative in the high-speed rapid chamber market, which is currently dominated by global leaders, to meet domestic industrial demands. ● Entry into High-Value Markets: Positioning the developed system as a strategic, high-value-added product in a market with high technical barriers and limited suppliers. ● Global Market Expansion: Leveraging the international standards and operational requirements integrated during development to secure technical specifications advantageous for overseas certification and long-term global export. **Industrial and Economic Impact** ● Technical Sovereignty & Supply Chain Stabilization: Mitigating foreign dependence for premium environmental test equipment, thereby reducing equipment acquisition/maintenance costs and minimizing supply chain risks. ● Strengthening Competitiveness of Key Domestic Industries: Contributing to shorter product development cycles and improved quality for the semiconductor, display, and automotive electronics sectors by providing stable, high-reliability testing infrastructure. ● Expansion of Premium Equipment and Testing Services: Promoting the high-value transformation of the testing and evaluation industry by expanding beyond hardware supply into data-driven services and customized testing solutions. ● Elevating Global Standing through International R&D: Setting a successful precedent for industry-led international collaborative R&D, enhancing the global brand recognition of South Korean testing equipment technology.

6. References